

MICHAEL OKORONKWO

Backend Software Engineer

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PROFESSIONAL SUMMARY

Backend software engineer with ~2 years of hands-on, production experience building REST APIs, payment integrations, and async processing systems in Node.js and Python. Shipped features live for a 7-person startup team and independently architected a full-stack scientific computing platform integrating a job queue, containerized pipeline, and relational data model. Graduated with a BSc (Hons) Pharmacology, bringing a rare, verifiable cross-disciplinary edge — computational chemistry, biostatistics, and regulated-industry rigor — that is directly applicable to health-tech, biotech, and data-intensive backend roles.

TECHNICAL SKILLS

- **Languages:** JavaScript · TypeScript · Python · PHP
- **Backend:** Node.js · Express.js · FastAPI · REST API Design · Redis · BullMQ (async job queues)
- **Databases:** PostgreSQL · MongoDB · Prisma ORM
- **Frontend:** React · Next.js · HTML · CSS
- **Infrastructure & Tools:** Docker · Git/GitHub · Postman · CI-style deployment workflows (Vercel)
- **Scientific Computing:** AutoDock Vina · RDKit · OpenBabel · Meeko · IBM SPSS

WORK EXPERIENCE

Backend Engineer — OunjeFood (Food-tech Startup), Lagos, Nigeria

Sep – Dec 2025

- Designed and shipped 30+ RESTful API endpoints in Node.js/Express — including JWT authentication built from scratch, menu management, order processing, and Paystack payment integration — as one of 2 backend engineers on a 7-person team.
- Diagnosed and resolved a critical OTP delivery failure caused by a firewall/IP whitelist misconfiguration, restoring authentication for all users.
- Fixed an idempotency bug in the order system that was generating duplicate orders, improving checkout data integrity.
- Built an automated daily vendor payout system to settle vendor wallet balances end-of-day, eliminating a manual finance process.
- Implemented upload image compression, directly reducing Cloudinary storage costs.

Quality Assurance Intern — Emzor Pharmaceuticals, Lagos, Nigeria

Jul – Sep 2025

- Supported QA processes for pharmaceutical manufacturing under strict regulatory compliance standards.
- Built analytical documentation habits from a highly regulated industry — directly transferable to writing precise specs and well-tested software.

PROJECTS

Virtual Drug Screening & Molecular Docking Microservice

Node.js/Express · Python/FastAPI · PostgreSQL · Redis/BullMQ · Docker · AutoDock Vina · RDKit · OpenBabel · Meeko

- Architected a full-stack virtual screening pipeline pairing a Node.js/Express API gateway with a Python/FastAPI compute service, using Redis/BullMQ to queue and process long-running docking jobs asynchronously (~20s avg runtime per job).
- Integrated AutoDock Vina, RDKit, OpenBabel, and Meeko into a containerized worker pipeline for ligand/receptor preparation and binding-affinity scoring.
- Designed a PostgreSQL schema tracking screening jobs, ligand libraries, and docking results for reproducible, queryable experiment history; containerized the full stack with Docker, independently debugging a WSL2-specific iptables/nf_tables kernel incompatibility.
- Directly applies pharmacology domain knowledge to computational drug discovery tooling — live demo below.

[Live demo: molecular-docking-frontend.vercel.app](#)

[Repos: github.com/Michael59-hub/molecular-docking-frontend & molecular-docking-microservice](#)

Buoyant — E-Commerce Platform

Next.js · TypeScript · Prisma ORM · PostgreSQL

- Built a full-stack e-commerce application end-to-end — PostgreSQL schema design via Prisma ORM, server-side rendering, and product/inventory management — using Next.js App Router with TypeScript for type-safe components.

[Repo: github.com/Michael59-hub/Buoyant](#)

EDUCATION

BSc (Hons) Pharmacology

Jul 2026

Olabisi Onabanjo University, Ago-Iwoye, Nigeria

- Completed Pfizer "Molecule to Medicine" industry-simulation programme. Coursework: Biochemistry, Biostatistics, Computational Chemistry.